Exam Paper Time-Lock: Initiation document · Case 03

Following chapter 1 (Initiation) of the Magshimim project-book template.

Description

A system that protects a national exam paper from leaks in two layers. **Lock:** the paper is encrypted and the opening key is split among five independent custodians so that only three together can open it, and only at the exam hour. **Trace:** every exam centre receives a copy with its own hidden watermark, so a leaked page names the centre it came from.

Instead of cancelling an exam for two million young people because "the paper leaked from somewhere", early opening is prevented and a leak points to an address.

Goal

The cancelled exam was the spark of the whole protest. The paper passes through many hands: setting, printing, packing, trucks, strongrooms and centres. When it leaks, nobody knows where, so the punishment is collective. Goal: make the secret something no single person holds, and a leak an address instead of a mystery.

Audience

- **Primary:** the National Testing Agency (NTA) and state exam boards.
- **Secondary:** centre managers who need a simple tool that opens the paper at the right time.
- **Scenario:** Madhuri Rao (a fictional character) runs a centre in Pune. On exam morning three of five key-holders approve on their phones and the paper opens in her centre at 14:00 sharp.

Existing solutions

- **CBSE, 2018:** after a leak, CBSE tried encrypted papers printed at centres shortly before the exam; centres reported delays and the approach was kept mainly as a backup. Feasible, and printing is the bottleneck.
- **Computer-based tests:** exams such as JEE Main already run on computers in controlled centres, with questions unlocked at start time.
- **Shamir secret sharing (1979):** standard threshold key splitting.
- **Printer tracking dots:** many laser printers embed hidden marks identifying the printer; covert marking on paper is possible.

What is new: a time-bound split key combined with a per-centre watermark, tuned for an exam of millions on one day.

Technological core

Threshold splitting (3 of 5) of the encryption key, and a per-centre watermark carried in the page layout (spacing and line breaks) that survives a photograph.

Main technologies

AES-256-GCM; Shamir secret sharing in a prime field; HSMs or smart cards for custodians; a layout engine (LaTeX/ReportLab) with per-centre layout variants. Open questions: pre-printing versus centre printing; layout watermark versus wording watermark (wording survives retyping but changes the question, a fairness problem).

Challenges and sources

- **Watermark robustness:** does it survive a photo? retyping? a leak in the form of an answer key? This is the research question.
- **The print window:** if the paper must be printed hours ahead, the lock protects only the file.
- **Custodian availability:** three of five must be reachable on time.

Sources: Shamir, "How to Share a Secret" (1979); reports on the CBSE encrypted-paper trial (2018); NIST SP 800-38D.

Hardware

None for the PoC. Deployment: HSMs for custodians; printers at centres if printing locally.

Conclusion

✅ Survives, on condition that the lab proves two claims: the watermark survives the way leaks actually spread, and the lock also protects the printed paper.